HUMAN GEOGRAPHY

Human geography is the study of the relationship between people and their environment. It examines how human activities shape landscapes, cultures, economies, and societies. This includes the study of population distribution, migration, urbanization, and the use of natural resources. Understanding these aspects is important for effective planning, sustainable development, and promoting international cooperation to create a balanced and inclusive world.

Naturalization of Humans and Humanization of Nature

The relationship between humans and nature has evolved over time, shaping the way societies interact with their environment.

- **Naturalization of Humans (Environmental Determinism)**
  - Early human societies were dependent on nature due to low technological development.
  - Example: Tribes like Benda's in central India rely on shifting cultivation, worship nature, and live in harmony with forests.
- **Humanization of Nature (Possibilism)**
  - With technological progress, humans modify nature for their needs.
  - Example: Use of genetic research for disease control, urbanization, etc.
- **Transition from Necessity to Freedom**
  - Human knowledge enables transformation of landscape fields, ports, cities, and space technology.
  - Nature becomes a resource for human activities, leading to cultural landscapes.

Conclusion: Humans first adapted to nature (determinism) but later modified it (possibilism), marking cultural advancement.

Neo determinism (Stop and Go Determinism)

It suggests that while nature sets certain limitations, human innovation can modify and adapt within these constraints. This approach views human-environment interaction as a controlled process, where technological advancements must align with ecological sustainability.

- **Concept by Griffith Taylor**
  - A middle path between **environmental determinism** (nature controls humans) and **possibilism** (humans control nature).
  - Example: Traffic lights regulate movement—humans modify nature but within limits.
- **Key Idea**
  - Humans **conquer nature by obeying it**, progressing when conditions allow.
  - Unregulated exploitation leads to **climate change, ozone depletion, and environmental degradation**.
- **Significance**
  - Encourages **sustainable development**, balancing growth with environmental responsibility.

Schools of Thought in Human Geography

- **Welfare/Humanistic Approach**
  - Focus: Social well-being (housing, health, education).
  - Example: *Geography of Social Well-being* introduced in academia.
- **Radical Approach**
  - Based on *Marxian theory*, linking poverty and inequality to capitalism.
- **Behavioral Approach**
  - Emphasizes *lived experiences* and *perception of space* based on ethnicity, race, and religion.

POPULATION

- The **population** of the world is unevenly distributed with some places having low concentrations of population and some having high. This is known as the **distribution of population.**
- According to the United Nations, the worldwide human population grew to **8.0 billion in mid-November 2022**, up from an estimated 2.5 billion in 1950. The world's population is predicted to grow by roughly **2 billion** during the **next 30 years.**
- **Key terms related to population:**

Term	**Explanation**
Crude Birth Rate	The annual number of live births per 1,000 people.
General Fertility Rate	The annual number of live births per 1,000 women of childbearing age (often taken to be from 15 to 49 years, but sometimes from 15 to 44).
Age-Specific Fertility Rates	The annual number of live births per 1,000 women in particular age groups (usually 15-19, 20-24, and so on).
Crude Death Rate	The annual number of deaths per 1,000 people.
Infant Mortality Rate	The annual number of deaths of children of age less than 1-year-old per 1,000 live births.
Life Expectancy	The number of years an individual at a given age can expect to live at present mortality levels. Life expectancy of India is 69.16 years (2017).
Total Fertility Rate (TFR)	The number of live births per woman completing her reproductive life, if her childbearing at each age reflected the current age-specific fertility rates.
Gross Reproduction Rate	The number of daughters who would be born to a woman completing her reproductive life at current age-specific fertility rates.
Net Reproduction Rate	The number of daughters who would be born to a woman according to current age-specific fertility and mortality rates.
Maternal Mortality Rate (MMR)	The number of maternal deaths per 100,000 live births due to pregnancy or termination of pregnancy, regardless of the site or duration of pregnancy.
Population Pyramid	A graphical illustration that shows the distribution of various age groups in a population (typically that of a country or region), forming a pyramid shape when the population is growing.
Sex Ratio	The number of females per thousand males.
Child Mortality Rate (CMR)	The number of child deaths under the age of 5 years per 1,000 live births. CMR in 2016: ~50; SDG Target for 2030: 11.
Dependency Ratio	Measure of the number of dependents aged 0-14 and over 65, compared with the total population aged 15-64.
Demographic Window	The period in a nation's demographic evolution when the proportion of the working-age population is particularly prominent.
Demographic Dividend	The economic growth potential results from shifts in a population's age structure, mainly when the share of the working-age population (15-64) is larger than the non-working-age population (0-14 and 65+).

Factors Influencing Population Distribution

I. Geographical Factors

- **Water Availability** – Essential for life, agriculture, industries, and navigation.
  - *Example: River valleys are densely populated*.
- **Landforms** – Flat plains support agriculture and infrastructure; mountainous areas hinder development.
  - *Example: Ganga plains vs. Himalayas*.
- **Climate** – Moderate climates attract people, while extreme conditions deter habitation.
  - *Example: Mediterranean regions vs. deserts*.
- **Soils** – Fertile soils support agriculture, leading to dense populations.
  - *Example: Indo-Gangetic plains vs. desert regions*.

II. Economic Factors

- **Minerals** – Mining and industries attract workers. *Example: Katanga-Zambia copper belt*.
- **Urbanization** – Cities offer jobs, education, healthcare, and better infrastructure, attracting migrants.
  - *Example: Growth of mega cities globally.*
- **Industrialization** – Industrial hubs create job opportunities, attracting diverse workers and service providers.
  - *Example: Kobe-Osaka region in Japan*.

III. Social and Cultural Factors

- **Religious & Cultural Significance** – Sacred places attract pilgrims and settlers.
  - *Example: Varanasi, Mecca*.
- **Political & Social Stability** – Conflict zones see depopulation, while peaceful areas attract migrants.
  - *Example: Migration from war zones*.
- **Government Policies** – Incentives for settling in underpopulated areas or relocating from overcrowded cities.
  - *Example: Settlement schemes in sparsely populated regions*.

Population Growth:

Growth of population refers to the change in the number of people living in a particular area between two points on time. Its rate is expressed in percentage.

- Population growth depends on birth rate, death rate and migration. The difference between fertility and mortality is called the **natural growth rate.**
  - ➢ This remarkable development has been fueled mostly by an increase in the number of individuals living to
    - ✓ **reproductive age**,
    - ✓ a gradual increase in human lifespan due to better medical facilities,
    - ✓ increased **urbanization**, and
    - ✓ accelerated **migration.**

Population Growth in India

The growth rate of population in India over the last century has been influenced by birth rates, death rates, and migration, showing different trends over time.

Types of Growth Rate

Growth Type	Explanation
Natural Growth Rate	The difference between crude birth rates and death rates between two points of time.
Induced Growth Rate	Determined by the volume of inward and outward movement of people in any given area.

Phases of population growth in India:

Phase	Period	Characteristics
Phase I	1901-1921	• Stagnant or stationary phase with extremely low or even negative growth (1911-1921). • High birth and death rates kept the population growth minimal. • Factors: Poor health services, illiteracy, and inefficient food distribution.
Phase II	1921-1951	• Period of steady population growth. • 1921 is known as the *Demographic Divide* due to a shift from zero to positive growth. • Improvement in health and sanitation reduced the mortality rate. • Crude birth rate remained high, leading to higher growth.
Phase III	1951-1981	• Population explosion phase. • Rapid decline in mortality but high fertility rate led to a sharp rise in population. • Developmental activities have improved living conditions. • Increased international migration (Tibetans, Bangladeshis, Nepalis, and Pakistanis) contributed to growth.
Phase IV	1981-Present	• The growth rate remains high but is gradually slowing down.

Migration: Causes & Types

Migration is the movement of people from one place to another, altering population distribution. It occurs due to economic, social, political, and environmental factors.

Types of Migration: Migration can be categorized based on duration and pattern:

1. **Based on Duration**
   - **Permanent Migration:** People relocate permanently, often due to employment, better living conditions, or conflict (e.g., rural-to-urban migration for jobs).
   - **Temporary Migration:** Movement for a limited period, such as labor migration for seasonal work.
   - **Seasonal Migration:** Migrants move temporarily due to climatic or economic reasons, like agricultural laborers moving for harvest seasons.
2. **Based on Pattern**
   - **Rural to Rural:** Movement within rural areas, often due to agricultural employment.
   - **Rural to Urban:** Most common type, driven by job opportunities, better education, and healthcare.
   - **Urban to Urban:** People shift from one city to another for better career prospects or lifestyle improvements.
   - **Urban to Rural:** Less common but it occurs when people seek a peaceful environment or lower cost of living.

Causes of Migration: Migration is driven by **push** and **pull** factors:

Push Factors (Forcing People to Leave Origin)	Pull Factors (Attracting Destination)
• **Economic Factors** ➢ Unemployment or low wages push people towards regions with better job prospects. ➢ Rural poverty forces people to migrate to urban areas for better income. • **Social Factors** ➢ Poor healthcare, lack of education, and low quality of life lead to migration. ➢ Ethnic conflicts and discrimination push communities to relocate. • **Political Factors** ➢ War, political instability, and oppressive regimes cause forced migration (e.g., refugees fleeing conflict zones). ➢ Persecution based on religion, race, or political beliefs forces people to migrate. • **Environmental Factors** ➢ Natural disasters (earthquakes, floods, droughts) displace people. ➢ Climate change and land degradation reduce agricultural productivity, pushing people to urban areas.	• **Economic Opportunities** ➢ Higher wages, industrialization, and business growth attract migrants. ➢ Urban centers provide better employment options compared to rural areas. • **Better Living Conditions** ➢ Improved healthcare, education, and infrastructure attract migrants. ➢ Cities offer better access to modern amenities, entertainment, and transportation. • **Political Stability & Security** ➢ A peaceful environment, strong governance, and legal protection attract migrants. ➢ Countries with strong democratic institutions attract skilled professionals. • **Favorable Climate** ➢ Regions with a moderate climate attract people from extreme climatic zones. ➢ Coastal cities and areas with pleasant weather often experience high migration rates.

Conclusion: Migration is a dynamic process influenced by economic, social, political, and environmental factors. While it offers opportunities for economic growth and cultural exchange, it also creates challenges like urban congestion, pressure on resources, and social integration issues. Understanding migration patterns helps policymakers manage population shifts effectively.

Demographic Transition

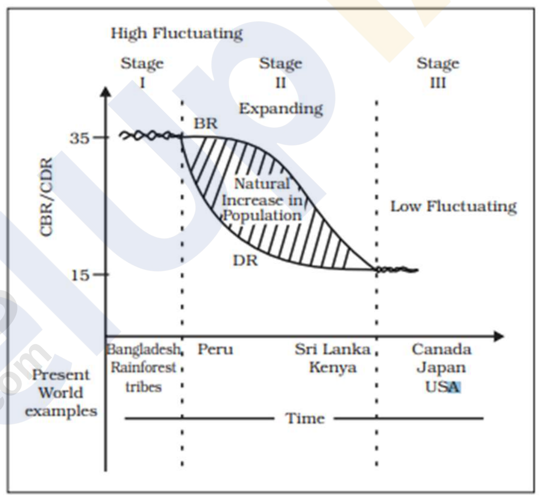
Demographic transition theory explains how population growth shifts as societies develop. It predicts population changes from **high birth and death rates** to **low birth and death rates** as a society evolves from a **rural, agrarian, and illiterate** structure to an **urban, industrial, and literate** one.

Stages of Demographic Cycle:

1. **High Birth & Death Rates:** Pre-industrial society, slow population growth.
2. **Declining Death Rates:** Improved healthcare, sanitation, rapid population growth.
3. **Declining Birth Rates:** Urbanization, education, slower growth.
4. **Low Birth & Death Rates:** Stabilized population.
5. **Declining Population (Optional):** Seen in some developed countries.

Stages of Demographic Transition

- **Stage 1: High Birth & Death Rates**
  - ➢ People reproduce more to compensate for high mortality.
  - ➢ Deaths occur due to epidemics and food scarcity.
  - ➢ Most people engage in agriculture; large families are an asset.
  - ➢ Life expectancy is low, and literacy levels are poor.
  - ➢ All countries were at these stage 200 years ago.



Three-staged model of Demographic Transition Theory

- **Stage 2: Declining Death Rate, High Birth Rate**
  - ➢ Mortality declines due to better sanitation and healthcare.
  - ➢ Birth rates remain high initially but start to decline gradually.
  - ➢ Population growth is accelerating due to the gap between births and deaths.

- **Stage 3: Low Birth & Death Rates**
  - ➢ Urbanization, literacy, and technological advancement increase.
  - ➢ People adopt family planning, reducing fertility rates.
  - ➢ Population stabilizes or grows at a slow pace.

Different countries today are at varying stages of demographic transition based on their socio-economic development.

Age Structure

Age structure refers to the distribution of people across different age groups within a population. It is a key indicator of population composition:

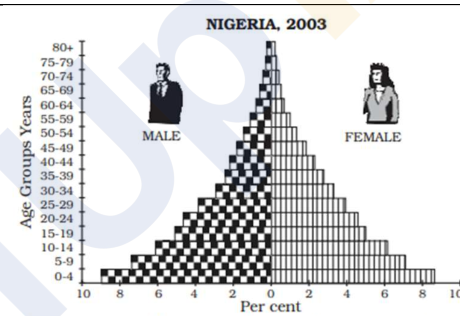
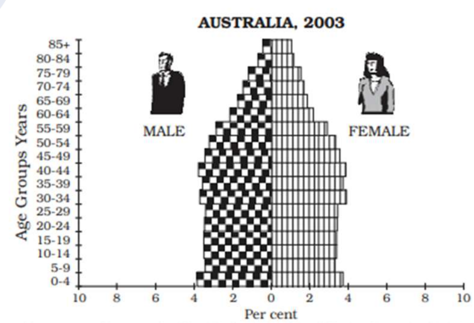
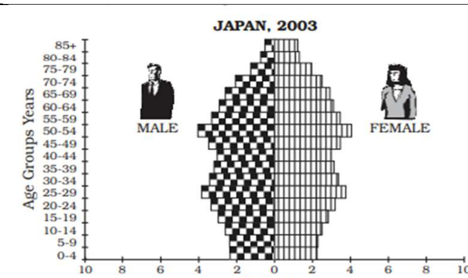
- A large proportion of people aged **15-59** signifies a strong **working population**.
- A higher percentage of individuals **above 60** indicates an **ageing population**, increasing the demand for healthcare and social support.
- A significant **young population** suggests a **high birth rate**, leading to a youthful demographic profile.

Age-Sex Pyramid

The **age-sex structure** of a population represents the number of males and females across different age groups. A **population pyramid** visually depicts this distribution:

- **Structure:** The left side shows **males**, and the right side shows **females** in each age group.
- **Shape Significance:** The pyramid's shape reflects population characteristics and growth trends.

Types of Population Pyramids:

1. **Expanding Population** (e.g., Nigeria) o **Shape:** Triangular with a **wide base** o **Features:** High birth rate, large young population o **Example:** Similar patterns in Bangladesh and Mexico	 *Data source: Demographic Year Book, 2003, United Nations Statistics Division. Data refer to national projection*
2. **Constant Population** (e.g., Australia) o **Shape: Bell-shaped**, tapering at the top o **Features:** Birth and death rates are nearly equal, leading to population stability	 *Data source: Demographic Year Book, 2003, United Nations Statistics Division.*
3. **Declining Population** (e.g., Japan) o **Shape: Narrow base**, tapered top o **Features:** Low birth and death rates, leading to zero or negative population growth	 *Data source: Demographic Year Book, 2003, United Nations Statistics Division. Excluding diplomatic personnel outside the country and foreign military and civilian personnel and their dependants stationed in the area*

Malthusian Theory of Population:

Thomas Robert Malthus was the first economist to propose a **systematic theory of population in 1798**. In his 1798 book *An Essay on the Principle of Population* he argued that **population growth tends to outpace food production**, leading to shortages, poverty, and social distress.

Key Assumptions of Malthusian Theory

- **Population Growth:** Population increases in a **geometric progression** (1, 2, 4, 8, 16...), meaning it doubles every 25 years if unchecked.
- **Food Supply Growth:** Food production grows in an **arithmetic progression** (1, 2, 3, 4, 5...), meaning it increases at a slower rate than population.
- **Resulting Crisis:** Since population grows faster than food supply, this imbalance leads to **famine, disease, and conflict**, reducing the population to sustainable levels.

Checks on Population Growth

To prevent excessive population growth, Malthus proposed two types of checks:

- **Positive Checks (Natural Calamities):** These increase the death rate and reduce population, including:
  - Famine
  - Wars and conflicts
  - Epidemics and diseases
  - Natural disasters
- **Preventive Checks (Human-Controlled Measures):** These reduce the birth rate, including:
  - Delayed marriages
  - Moral restraints (abstinence)
  - Family planning and contraception (though Malthus was skeptical of artificial birth control)

Criticism of Malthusian Theory

- **Underestimation of Technological Advancements:** The Green Revolution and modern agriculture have disproved Malthus's claim that food supply grows only arithmetically.
- **Declining Fertility Rates:** Many countries today experience low birth rates due to education, urbanization, and economic development.
- **Economic Growth and Industrialization:** Improved living standards and better resource management prevent crises predicted by Malthus.

Relevance Today

While Malthus's predictions have not come true globally, his theory remains relevant in regions facing **overpopulation, food shortages, and resource depletion**, such as parts of Africa and South Asia. The concept has also influenced neo-Malthusian ideas on **sustainability, climate change, and population control policies**.

Population in India: A Challenge and Potential Opportunity

- The current population of India is at **1,441.7** according to UNFPA data.
- Population of India is equivalent to 17.7% of the total world population. India ranks number 1 in the list of countries (and dependencies) by population.
- According to estimates in a recently released United Nations report, Indias' population is expected to reach 1.7 billion by the year 2050.
- Decadal Growth rate of Population in India between 2001-2011 was 17.64%. It decreased from 21.54% during 1991-2001.

Challenges of Population Growth in India

- **Unemployment and Underemployment**
  - High population growth increases job competition, leading to **youth unemployment** and **informal sector growth**.
  - Over 40% of India's workforce is engaged in agriculture, often underemployed.
- **Pressure on Natural Resources**
  - Overuse of land, water, and forests leads to **deforestation, groundwater depletion, and pollution**.
  - Per capita availability of water and arable land is declining.
- **Urbanization and Infrastructure Deficit**
  - Rapid urban migration results in **slums, traffic congestion, and inadequate housing**.
  - Cities like Delhi and Mumbai face **water shortages and air pollution**.
- **Health and Education Burden**
  - High population strains **healthcare and education systems**.
  - Malnutrition and disease outbreaks remain concerns.
- **Food Security and Inflation**
  - Rising demand for food increases **inflation and dependence on imports**.
  - Climate change affects agricultural productivity.

Measures taken by government:

- **Population Stabilization Measures**
  - **National Population Policy (NPP), 2000** – Aims to **achieve replacement-level fertility (TFR of 2.1)** and promote **family planning awareness** – So, far this policy successfully achieved TFR of 2.0, showing a declining trend in population growth rate.
  - **Mission Parivar Vikas (2016)** – Focuses on high-fertility districts in **Bihar, UP, MP, Rajasthan, Jharkhand, Chhattisgarh, and Assam** to improve access to contraception and reproductive health services.
  - **National Family Planning Program** – Provides free contraceptives, sterilization, and counseling through government healthcare facilities.

- **Skill Development and Employment**
  - **Skill India Mission** – Aims to train **400 million youth** in vocational skills by 2025.
  - **PM Kaushal Vikas Yojana (PMKVY)** – Provides short-term training and certification in job-oriented skills.
  - **Startup India and Standup India** – Encourages entrepreneurship and self-employment among youth and women.

- **Education and Human Capital Development**
  - **Right to Education (RTE) Act, 2009** – Ensures **free and compulsory education** for children aged 6-14 years.

- **Samagra Shiksha Abhiyan** – Integrates primary, secondary, and higher education to improve learning outcomes.
- **National Education Policy (NEP), 2020** – Focuses on **skill-based education, vocational training, and digital learning**.

- **Healthcare and Nutrition**
  - **Ayushman Bharat (PM-JAY)** – Provides **free health insurance** to over **10 crore vulnerable families**.
  - **Poshan Abhiyaan** – Targets **malnutrition, stunting, and anemia** among children and women.
  - **National Health Mission (NHM)** – Strengthens **maternal and child healthcare services**.

- **Urbanization and Infrastructure Development**
  - **Smart Cities Mission** – Aims to develop **100 smart cities** with improved urban planning and infrastructure.
  - **PM Awas Yojana (Urban & Rural)** – Provides **affordable housing** for urban and rural poor.
  - **AMRUT (Atal Mission for Rejuvenation and Urban Transformation)** – Focuses on **urban sanitation, water supply, and mobility**.

- **Agricultural and Food Security Measures**
  - **PM Kisan Samman Nidhi** – Provides **₹6,000 per year** to small and marginal farmers.
  - **National Food Security Act (NFSA), 2013** – Ensures subsidized food grains for **75% of rural and 50% of urban population**.
  - **Doubling Farmers' Income Initiative** – Aims to improve **agricultural productivity and market access**.

Way Forward

- **Strengthening Family Planning Awareness –** Expanding access to contraception and reproductive healthcare.
- **Enhancing Education and Skill Training –** Making education more employment-oriented.
- **Sustainable Urban Development –** Improving housing, sanitation, and transport in cities.
- **Expanding Healthcare Services –** Strengthening rural healthcare infrastructure.

India's population is both a **challenge and an opportunity**. With the right policies, **it can transform into a driving force for economic growth and global leadership**.